Exercise - 2.9

Resolve the following rational expressions into partial fractions.

1. $\dfrac{1}{x^2 - a^2}$
2. $\dfrac{3x+1}{(x-2)(x+1)}$
3. $\dfrac{x}{(x^2+1)(x-1)(x+2)}$
4. $\dfrac{x}{(x-1)^3}$
5. $\dfrac{1}{x^4 - 1}$
6. $\dfrac{(x-1)^2}{x^3 + x}$
7. $\dfrac{x^2+x+1}{x^2 - 5x + 6}$
8. $\dfrac{x^3 + 2x + 1}{x^2 + 5x + 6}$
9. $\dfrac{x+12}{(x+1)^2(x-2)}$
10. $\dfrac{6x^2 - x + 1}{x^3 + x^2 + x + 1}$
11. $\dfrac{2x^2 + 5x - 11}{x^2 + 2x - 3}$
12. $\dfrac{7+x}{(1+x)(1+x^2)}$

2.7.3 Graphical Representation of Linear Inequalities

A straight line $ax + by = c$ divides the Cartesian plane into two parts. Each part is an half plane. A vertical line $x = c$ will divide the plane in left and right half planes and a horizontal line $y = k$ will divide the plane into upper and lower half planes.

A point in the cartesian plane which is not on the line $ax + by = c$ will lie in exactly one of the two half planes determined by the line and satisfies one of the inequalities $ax + by < c$ or $ax + by > c$.

To identify the half plane represented by $ax + by < c$, choose a point P in any one of the half planes and substitute the coordinates of P in the inequality.

If the inequality is satisfied, then the required half plane is the one that contains P; otherwise the required half plane is the one that does not contain P. When $c \neq 0$, it is most convenient to take P to be the origin.

Example 2.28 Shade the region given by the inequality $x \geq 2$.

Solution:
First we consider equation $x = 2$. It is a line parallel to y axis at a distance of 2 units from it. This line divides the cartesian plane into two parts. Substituting $(0, 0)$ in the inequality we get $0 \geq 2$ which is false. Hence the region which does not contain the origin is represented by the inequality $x \geq 2$. The shaded region is the required solution set of the given inequality. Since $x \geq 2$, the points on the line $x = 2$ are also solutions.

